Team Name	CSE4104 - 7B - T04
Section	7B
Project Title	AI Programming Tutor & Bug Explainer
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GitHub Repository	https://github.com/Rayhan-55/cse4104-7b-t04-ai-programming-tutor-and-Bug-Explainer

Project Title: AI Programming Tutor & Bug Explainer

Preliminary Project Idea

AI Programming Tutor & Bug Explainer is an AI-based educational software project designed to help beginner programmers learn coding more easily and solve programming errors efficiently. The system acts like a virtual programming mentor that provides instant explanations, debugging support, and learning guidance. The main goal of the project is to reduce the difficulties students face while learning programming languages, understanding compiler errors, and debugging code.

The platform will help students by:

- Explaining coding concepts in simple language
- Detecting and explaining programming errors
- Suggesting code improvements
- Helping users understand GitHub projects
- Providing personalized programming support
- Supporting learning through AI-powered conversations

The project mainly targets beginner programmers, CSE students, and self-learners.

Initial Technology Selection

Frontend Technologies

Technology	Purpose
React.js	Frontend development

Tailwind CSS	User interface styling
Axios	API communication
Monaco Editor	Online code editor
React Router	Navigation system

Backend Technologies

Technology	Purpose
Node.js	Server-side runtime
Express.js	Backend API development
MongoDB	Database management
Mongoose	Database modeling
JWT Authentication	Secure login system
bcrypt	Password encryption

AI Technologies

Technology	Purpose
OpenAI API / Gemini API	AI-based tutoring and explanation
Vision AI	Image and PDF understanding

Prompt Engineering	Beginner-friendly AI responses

Other Tools

Tool	Purpose
GitHub API	Repository analysis
GitHub	Version control
Browser Speech API	Voice assistant support

Preliminary Project Description

AI Programming Tutor & Bug Explainer is a smart educational platform that uses Artificial Intelligence to simplify programming education.

Many beginner programmers struggle with:

- Understanding programming logic
- Solving compiler and runtime errors
- Reading difficult documentation
- Understanding GitHub repositories
- Debugging code efficiently

This project aims to solve these problems by creating an AI-powered virtual tutor that explains programming topics in simple English, Bangla, or Banglish.

The system includes several intelligent modules:

1. AI Programming Tutor

Users can ask programming-related questions and receive beginner-friendly explanations with examples.

2. Bug Explainer

Users can paste source code and receive:

- Error detection
- Line-by-line explanation
- Suggested fixes
- Better coding practices

3. Compiler Error Translator

Complex compiler messages are converted into simple human-readable explanations.

Example:

“Null Pointer Exception” → “Your program is trying to use a variable that does not contain any valid value.”

4. GitHub Repository Analyzer

Users can submit a GitHub repository link and the AI explains:

- Project purpose
- Folder structure
- Technologies used
- Important files

5. PDF & Image Explainer

Users can upload:

- Lecture notes
- PDFs
- Screenshots
- Handwritten coding problems

The AI analyzes and explains the uploaded content.

6. Personalized Learning Support

The system tracks user activities and weak topics to provide personalized learning suggestions.

7. Voice Tutor

AI responses can be converted into voice for better accessibility.

Why AI is Needed in the Project

Artificial Intelligence is the core component of this project because traditional educational systems cannot provide instant personalized support for every programming student. Without AI, the platform would only work as a static tutorial website and would not be able to:

- Understand user questions dynamically
- Analyze programming code intelligently
- Explain compiler errors contextually
- Personalize learning experiences

- Analyze uploaded documents and images
- Provide instant tutoring support

AI plays several important roles in the project:

Intelligent Code Analysis

AI can understand source code and detect:

- Syntax errors
- Logical mistakes
- Runtime problems
- Optimization opportunities

Human-Friendly Error Explanation

AI converts difficult compiler messages into beginner-friendly explanations.

Personalized Learning Experience

AI tracks user weak areas and adapts future explanations accordingly.

Natural Language Communication

Users can communicate using English, Bangla, or Banglish.

Smart Repository & Document Understanding

AI can summarize and explain:

- GitHub repositories
- PDFs

- Notes
- Images
- Handwritten problems

24/7 Learning Support

Unlike human tutors, AI can provide continuous support anytime.

Scalability

One AI system can support thousands of students simultaneously.

AI Programming Tutor & Bug Explainer is an intelligent educational software project that aims to simplify programming education with the help of Artificial Intelligence. The project is designed especially for beginner programmers and students who often face difficulties understanding programming concepts, debugging code, and solving compiler errors. By combining AI-powered tutoring, intelligent code analysis, compiler error explanation, GitHub repository understanding, personalized learning support, and voice assistance, the platform creates a complete smart learning environment for users. The uploaded project already contains a strong technical foundation with modern technologies and multiple useful features, making it a highly promising academic as well as real-world educational software solution.